

Paper Number: P2075R4
Title: Philox as an extension of the C++ RNG engines
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Audience: LEWG
Date: 2024-02-14

1. Introduction

C++11 introduced a comprehensive mechanism to manage the generation of random numbers in the `<random>` header file (including distributions, pseudo random and non-deterministic engines).

We proposed a set of engine candidates for the C++ standard extension in the P1932R0 paper [1]. This paper is focused on the family of the counter-based Philox engines.

2. Revision History

Key changes compared with R3 (reviewed at telecon 2023-11-28 in LEWG):

- Dropped `counter_based_engine`-based approach. The remaining one is Philox-specific..
- Moved relevant API restrictions into Mandates sections.
- Aligned synopsis with other engines (added streaming operator)
- Added actual 10000th values into corresponding section

Key changes compared with R2 (reviewed at telecon 2023-09-26 in LEWG):

- Dropped several aliases for simplification.
- changed API for `std::counter` to control number of values passed
- extended design consideration section to discuss `set_counter` and `get_counter` functionality and API
- added explicit mention of `little_endian` notation for interpretation of the words passed to `set_counter` in order to keep code portability
- extended `pseudo_random_function` concept with `counter_size` parameter to avoid ambiguous logic of calculation of this value, previously `counter_size` assumed to be equal to `output_count`, which is not necessarily the case in all cases

Key changes compared with R1 (reviewed at telecon 2022-05-22 in SG6):

- Wording for the Philox-focused API was simplified.
- Wording for the `counter_based_engine` based API was extended.
- Design considerations section was added.
- `set_counter()` member function was added to the engine.
- `c` template parameter was removed for the sake of ease of use.

Key changes compared with R0 (reviewed in Prague in SG6):

- Aligned wording for `philox_engine` with the C++ standard.
- Added an alternative API with a `std::array` template parameter. Removed alternative APIs with calculated constant values.
- Added an alternative approach with a generic `counter_based_engine` and a specific `philox_prf` pseudo-random function.

3. Motivation

Random number generators (engines) are at the heart of Monte Carlo simulations used in many applications such as physics simulations, finance, statistical sampling, cryptography, noise generation and others.

Each of the C++11 random number generators has own advantages and disadvantages, e.g. linear congruential generators, the simplest generators with 32-bit state, has a quite short generation period (2^{32}) and weak statistical properties, while Mersenne Twister 19937 generator has long generation period and strong statistical properties, but has a large vector state that affects efficiency of parallelism in Monte Carlo simulations.

Several new algorithms were introduced in the last decade, which can utilize modern hardware parallelism and provide solid statistical properties.

4. General Description

Philox is one of the counter-based engines introduced in 2011 in [2]. All counter-based engines have a small state (e.g., Philox4x32 has 10 x 32-bit elements in its state) and a long period (e.g., the period of Philox4x32 is 2^{130}). Counter-based engines effectively support parallel simulations via both block-splitting and independent-stream techniques and many of them (including Philox) are well-suited to a wide variety of hardware including CPU/GPU/FPGA/etc.

Philox is proposed as the first new engine since C++11 for standardization. It satisfies the following criteria, as discussed in P1932R0 [1]):

- **Statistical properties.** The original paper asserted that the Philox family passes rigorous statistical tests including hundreds of different invocations of TestU01's BigCrush [2]. This statement has been independently confirmed: the TestU01 batteries for Philox4x32-10 and Philox4x32-7 were tested in [4] and DieHard testing results for Philox4x32-10 were published in the Intel® Math Kernel Library (Intel® MKL) documentation [5].
- **Wide usage.** Philox is broadly used in Monte-Carlo simulations which require massively parallel random number generation, e.g., financial simulations [6], simulation of non-deterministic finite automata [7], etc.
- **HW friendliness.** Philox's distinguishing features are its small state and reliance on simple primitive operations. It is, therefore, easy to vectorize and parallelize.

The value of the Philox engine is widely recognized by various vendors, to name a few:

- Intel* provides implementation of Philox4x32-10 as part of Intel® oneAPI Math Kernel Library.
- Nvidia* provides implementation of Philox4x32-10 as part of cuRAND library.
- AMD* provides implementation of Philox4x32-10 as part of rocRAND project.
- MathWorks* provides GPU-optimized implementation of Philox4x32-10 as part of their product.
- Microsoft* is using Philox4x32-10 as part of DirectML project.
- numPy provides implementation of Philox4x64-10.
- cuPy provides implementation of Philox4x32-10.

5. High-level API Design

The API defines a self-contained engine template class analogous to the other random number engines in the standard.

New engine introduces a dedicated function `.set_counter()` to set the state to arbitrary position, which enables support of parallel simulations and is a trivial operation. Its use cases are described in the Design considerations section.

6. Philox engine API and Wording

This API specifies a single, new `philox_engine` class template.

```
template<typename UIntType, std::size_t w, std::size_t n, std::size_t r, UIntType ...consts>
class philox_engine;
```

The `philox_engine` is described in terms of the Philox function which acts as a keyed bijection on a domain of size $2^{w \cdot n}$. Consequently, the `philoxNxW` engines have a period of $n \cdot 2^{w \cdot n}$.

Pre-defined aliases are provided for instantiations with constants and parameters that are known to produce high-quality random numbers.

The `philoxNxW` aliases have a pre-defined round-count, `r=10`, that is somewhat larger than the minimum required to pass known statistical tests, but is widely used in practice. In other words, they provide a statistical safety margin at a modest performance cost.

```
using philox4x32 = ...;
using philox4x64 = ...;
```

- **Wording**

The changes affect only section “Random number generation [rand.general]”.

- **Changes in section [rand.general]**

...

(5.3) – the operator `mullo` denotes the low half of the modular multiplication of `a` and `b`: $(a \cdot b) \bmod 2^w$

(5.4) – the operator `mulhi` denotes the high half of the multiplication of `a` and `b`: $\lfloor (a \cdot b) / 2^w \rfloor$

- **Changes in sub-section [rand.synopsis] Header <random> synopsis**

...

```
// [rand.eng.philox] class template philox_engine
```

```
template<typename UIntType, std::size_t w, std::size_t n, std::size_t r,
        UIntType ...consts>
    class philox_engine;
```

...

```
// [rand.predef] engines and engine adaptors with predefined parameters.
```

...

```
using philox4x32 = see below;
```

```
using philox4x64 = see below;
```

...

- New sub-section after [rand.eng.sub]

Class template **philox_engine** [rand.eng.philox]

- 1 A **philox_engine** random number engine produces unsigned integer random numbers in the closed interval $[0, 2^w - 1]$, where the template parameter w defines the range of the produced numbers. The state of a **philox_engine** object is of size $(5n/2+1)$ and consists of a sequence X of n **result_types**, a sequence K of $n/2$ **result_types**, a sequence Y of n **result_types**, and a scalar i , where:
 - X is a *counter*,
 - K are keys,
 - Y is a buffer of produced values,
 - and i is an index in Y buffer.
- 2 The generation algorithm returns Y_i , the value stored in the i^{th} element of Y after applying the state transition algorithm.
- 3 The state transition algorithm is performed as follows:

```

i=i+1
if (i == n) {
    Y = Philox(K, X) // see below
    X = (X+1)
    i = 0
}

```

where X behaves as if it is a $n \cdot w$ -bit integer

- 4 The Philox function maps the $n/2$ -length sequence K and the n -length sequence X into an n -length output sequence Y . Philox applies an r -round substitution-permutation network to the values in X . A single round of the generation algorithm performs the following steps:

(4.1) – The output sequence X' of the previous round (X in case of the first round) is permuted to obtain the intermediate state V :

$$V_j = X'_{f(j)}$$

where $j = 0, \dots, n - 1$ and $f(j)$ is defined in Table 1 below, as in [2, 9]:

Table 1. Values for the word permutation $f(j)$

		$j=$															
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$n=$	2	0	1														
	4	0	3	2	1												
	8	2	1	4	7	6	5	0	3								
	16	0	9	2	13	6	11	4	15	10	7	12	3	14	5	8	1

[Note: for $n=2$ the sequence is not permuted]

(4.2) – The following computations are applied to the elements of the V sequence:

$$X'_{2 \cdot k} = \text{mullo}(V_{2 \cdot k+1}, M_k)$$

$$X'_{2 \cdot k+1} = \text{mulhi}(V_{2 \cdot k+1}, M_k) \text{ xor } \text{key}_k^q \text{ xor } V_{2 \cdot k}$$

where: $k = 0 \dots n/2-1$ is the index in the sequences, q is the index of the round: $q = 0 \dots r - 1$, key_k^q is the k^{th} round key for round q , $key_k^q = (K_k + q \cdot C_k) \bmod 2^w$, K_k are the keys generated once with a seed and stay constant if not reseeded, M_k are multipliers (odd elements of template parameters `consts`) and C_k are `round_consts` (even elements of template parameters `consts`).

5 After r applications of the single-round function, Philox returns the sequence $Y = X'$.

```
template<typename UIntType, std::size_t w, std::size_t n, std::size_t r,
        UIntType ...consts>
class philox_engine {
    // Exposition only
    static constexpr std::size_t array_size = n / 2;

public:
    // types
    using result_type = UIntType;

    // engine characteristics
    static constexpr std::size_t word_size          = w;
    static constexpr std::size_t word_count         = n;
    static constexpr std::size_t round_count        = r;
    static constexpr std::array<result_type, array_size> multipliers;
    static constexpr std::array<result_type, array_size> round_consts;
    static constexpr result_type min() { return 0; }
    static constexpr result_type max() { return 2w - 1; }
    static constexpr result_type default_seed = 20111115u;

    // constructors and seeding functions
    philox_engine() : philox_engine(default_seed) {}
    explicit philox_engine(result_type value);
    template<class Sseq> explicit philox_engine(Sseq& q);
    void seed(result_type value = default_seed);
    template<class Sseq> void seed(Sseq& q);

    void set_counter(const std::array<result_type, n>& counter);

    // equality operators
    friend bool operator==(const philox_engine& x, const philox_engine& y);

    // generating functions
    result_type operator()();
    void discard(unsigned long long z);

    // inserters and extractors
    template<class charT, class traits>
        friend basic_ostream<charT, traits>&
            operator<<(basic_ostream<charT, traits>& os, const philox_engine& x);
    template<class charT, class traits>
        friend basic_istream<charT, traits>&
            operator>>(basic_istream<charT, traits>& is, philox_engine& x);
};
```

6 *Mandates:*

- `sizeof...(consts) == n` is true, and
- `n == 2 || n == 4 || n == 8 || n == 16` is true, and
- `0 < r` is true, and
- `w <= numeric_limits<UIntType>::digits` is true.

7 The template parameter `...consts` represents the M_k and C_k constants which are grouped as follows: $[M_0, C_0, M_1, C_1, M_2, C_2 \dots M_{N/2-1}, C_{N/2-1}]$

- 8 The textual representation consists of the values of $K_0, \dots, K_{n/2-1}, X_0, \dots, X_{n-1}, i$, in that order. Note that the stream extraction operator can reconstruct Y from K and X , as needed.

```
explicit philox_engine(result_type value);
```

- 9 *Effects:* Sets the K_0 element of sequence K to value. All elements of sequences X and K (except K_0) are set to 0. The value of i is set to $n-1$.

```
template<class Sseq> explicit philox_engine(Sseq& q);
```

- 10 *Effects:* With $W = \lceil w/32 \rceil$ and an array (or equivalent) a of length $(n/2) \cdot W$, invokes $q.generate(a + 0, a + n / 2 * w)$ and then iteratively for $i=0, \dots, n/2-1$, sets K_i to $\left(\sum_{j=0}^{W-1} a[i \cdot W + j] \cdot 2^{32 \cdot j} \right) \bmod 2^w$. All elements of sequence X are set to 0. The value of i is set to $n-1$.

```
void set_counter(const std::array<result_type, n>& counter);
```

- 11 *Effects:* Sets the X_i for $i = 0, \dots, n-1$ elements of sequence X starting from words containing the most significant parts of the counter value in the descending order to values `counter[i]`. The value of i is set to $n-1$.

- Changes in sub-section *[rand.predef]* Engines and engine adaptors with predefined parameters

...

```
using philox4x32 = philox_engine<uint_fast32_t, 32, 4, 10, 0xD2511F53, 0x9E3779B9,
0xCD9E8D57, 0xBB67AE85>;
```

- *Required behavior:* The 10000th consecutive invocation of a default-constructed object of type `philox4x32` produces the value 1955073260.

```
using philox4x64 = philox_engine<uint_fast64_t, 64, 4, 10, 0xD2E7470EE14C6C93,
0x9E3779B97F4A7C15, 0xCA5A826395121157, 0xBB67AE8584CAA73B>;
```

- *Required behavior:* The 10000th consecutive invocation of a default-constructed object of type `philox4x64` produces the value 3409172418970261260.

- Changes in *[version.syn]*

Add a predefined macro to *[version.syn]*:

```
#define __cpp_lib_philox_engine 202310L // also in <random>
```

8 Design considerations

1 Compare approaches

Two approaches to an API definition were considered:

1. A philox-focused API defines a self-contained engine class template analogous to the other random number engines in the standard. (This is an evolution of the R0 version of this paper).
2. A counter-based-engine API, which is more generic and allows the creation of engines based on other pseudo-random functions as well.

For more details see [R3 revision](#).

Based on the poll at LEWG Telecon on 2023-11-28 the second approach was removed from the paper starting from R4.

2 set_counter use case

The following example shows the typical flow for a Monte Carlo simulation of a large number of "atoms" for a large number of timesteps:

```
uint32_t global_seed = 999;
for(uint32_t time_step = 0; time_step < time_steps_num; ++time_step){
    for(uint32_t atom_id = 0; atom_id < atoms_num; ++atom_id){
        philox4x32 eng(global_seed);
        eng.set_counter({atom_id, time_step, 0, 0});
        normal_distribution nd;
        auto n1 = nd(eng);
        auto n2 = nd(eng);
        // ...
    }
}
```

Using `set_counter()` allows creation of the engine on the fly without storing `atoms_num` of states. In addition it does not prevent parallelisation of either of the loops.

On the down side, one should control the number of random numbers consumed per timestep per atom. If the number consumed numbers overcome $4 \cdot 2^{32 \cdot 2}$, then sequences in different atoms may overlap, which brings in undesired cross correlation. The following section discussed the way to avoid that.

Under certain limitations a similar effect can be achieved via using `.discard()` function, but it differs in several aspects. The most critical one:

- `.discard()` shifts are limited to *unsigned long long*, which on many systems is 64-bits integer, while `philox4x64` has a period of $4 \cdot 2^{64 \cdot 4}$, thus splitting this sequence in 2 parts would require $4 \cdot 2^{64 \cdot 3} - 64$ calls of `discard()`, while `.set_counter()` can do the same in one call.

There are other differences:

1. `.discard()` shifts the counter only forward relative to its current position. This API exists because some (but not all) engines have efficient algorithms to move their state forward.
2. `.set_counter()` sets the absolute value for the counter. It is a unique property of counter-based engines - it is trivial to set their absolute state.

3 set_counter API

[P2075R2](#) defined API as `void set_counter(std::initializer_list<result_type> counter)`, which allowed passing more than `n` (template argument of `philox_engine` class) required values, with excessive values silently ignored.

From the mathematical point counter is a single big integer described in wording sections as `X`. There is no existing facility in the standard, which can represent this entity in its mathematical sense. There is the paper P1889R1, which introduces `std::wide_integer<Bits, S>` that may potentially be useful in the future in different aspects of random numbers facilities, but it is targeted to Numerics TS and is not specific to `set_counter()`, thus we decided to not rely on this facility.

We considered 4 options to fix this API:

1. `void set_counter(std::span<const UIntType, n> counter);`
 - a. `obj.set_counter({{atom_id, time_step, 0, 0}});`

The most common use case for `set_counter` functionality is a temporary object created on the fly and not a real container. The main use case requires double braces, which might be not syntactically friendly for average users. This API enforces exactly `n` values passed, which addresses the main concern of the API from revision of the paper.

Double braces will not go away with P2447R4, because the newly introduced constructor for `std::span` is explicit for cases with non `dynamic_extent`.

```
2. void set_counter(const std::array<UIntType, n>& counter);  
    a. obj.set_counter({atom_id, time_step, 0, 0});
```

This approach is slightly less generic - it fixes `std::array` as the only acceptable container. The corner case of this approach - it allows passing less variables in curly braces than required with assumed zero-tail. Passing less value can be considered reasonable, because the sequence is being split for parallelization, the list significant bits are left for consumption within the block of computations and can be zeroed.

```
    b. obj.set_counter({atom_id, time_step});
```

Existing Philox aliases allow engines with `n=4`, which translates into 4 elements in the input array. But general facilities allow instantiation of the template with `n` up to 16. In this case, the way to avoid typing all of the 16 elements would be useful. That said, authors are not aware of `n>4` usage in real user codes.

Thus in ninja use cases that approach is less verbose but is slightly more error prone.

```
3. void set_counter(see below);  
    a. obj.set_counter(atom_id, time_step, 0, 0);
```

It is assumed, that description would define the signature, which requires exactly `n` of `UIntType` values to be passed to the function and that solves the main problem of APIs from previous revision of the paper. At the same time this approach is harder to understand from its specification and it loses perception that the counter is actually a single entity because of several arguments of the function

```
4. void set_counter(const counter_t& counter);  
    a. obj.set_counter({atom_id, time_step, 0, 0});
```

If `counter_t` covers representation of the counter entity in its aspects needed for `set_counter`, then it becomes a thin wrapper over `std::array`. If it is extended to cover additional manipulations with the counter it represents, then it needs additional arithmetic operations and becomes a thin wrapper around non-existing `std::wide_integer`. In both cases we see no strong reason to introduce a new type.

Within the provided set of cases, we decided to stick to:

```
void set_counter(const std::array<UIntType, n> counter);
```

It is less generic than `std::span`, but avoids verbosity of additional braces. When/if `std::wide_integer` becomes the standard, additional overload for `set_counter` can potentially be introduced if considered necessary.

4 `get_counter` member function

`get_counter()` function is a natural counterpart for the introduced `set_counter()` function. But it is important to note, that there are several gotchas, which makes it less convenient, than one may consider:

1. Counter is not fully represent the state of the engine. Looking into Wording section, especially the description of the transition algorithm TA, one can see that the counter is being ticked only

every n -th invocation of `operator()` of the engine. This happens because the Philox algorithm generates a batch of n values each time, thus we have a buffer Y , which stores $n-1$ values to return before the next run of the Philox algorithm. In order to fully restore the state of the Philox, one should restore the counter, buffer Y and the position i inside this buffer.

2. `set_counter` is useful, when we statically divide the whole period of the Philox generator into several sub-sequences and further do the whole computation within these subsequences. That approach does not require checking the current state of the engine.
3. Let us consider dynamic parallelism, when we get the engine in some unknown state and plan to parallelise some amount of work. This use case is covered by existing `.discard()` functionality, where engine state is being shifted in relative manner. If one wants to reproduce that scenario with `set/get_counter()`, additional arithmetics should be introduced for counter type, which would require efforts comparable with introduction of `std::wide_integer<>` discussed in section about `set_counter` API.

Within the provided set of considerations, we decided to not introduce `get_counter()`. One can return to this question when additional use cases are found and/or `std::wide_integer<>` be accepted for the standard.

5 Splitting sequence in sub sequences

[P2075R1](#) revision of this paper had `c` template argument for *counter_based_engine*:

```
template<pseudo_random_function prf, size_t c>
class counter_based_engine.
```

The main purpose of this parameter was to split counter X into lower c words X_1 and higher $n-c$ words X_2 . X_1 behaves as a normal counter and wraps when depleted. X_2 is predefined by the user and is considered constant by the algorithm.

The intention of this parameter was to add a simple way to split a full sequence of random numbers into independent $(n-c)*word_size$ subsequences, which can be used for parallelisation and easy creation of such subsequences on the flight.

Further analysis revealed that this concept can be applicable for a wider set of engines, which makes `c` parameter on the level of the engine not generic enough.

As a further design consideration for this methodology we propose to consider a dedicated additional adapter, such as:

```
template<template Engine, size_t c>
class subsequence_engine;
```

This adaptor can be customized for a subset of engines where dedicated optimizations are possible.

Further investigations for this adaptor can be done in a separate paper. Authors removed the `c` parameter from this revision.

6 Using `std::array` in template arguments

The template parameter `consts` as a `std::array` was considered.

```
// *****
// Alternative API: consts template parameter represented as std::array
// *****

template<typename UIntType, std::size_t w, std::size_t n, std::size_t r,
std::array<UIntType, n> consts>
class philox_engine {
    static constexpr std::size_t array_size = n / 2; // Exposition only
```

```
public:
...
    static constexpr std::array<result_type, array_size> multipliers;
    static constexpr std::array<result_type, array_size> round_consts;
...
```

`philox_engine` class template is not expected to be frequently used by users - predefined aliases are the main way to use this engine. Having that in mind, we decided to not introduce a new API technique into standard for a minor simplification.

7 Additional aliases `philox2x32` and `philox2x64`

Original paper contained additional aliases:

```
template<size_t r>
using philox2x32_r = philox_engine<uint_fast32_t, 32, 2, r, 0xD2511F53, 0x9E3779B9>;
```

- 1 Required behavior:** The 10000th consecutive invocation of a default-constructed object of type `philox2x32_r<10>` produces the value XXXXXXXXX

```
template<size_t r>
using philox2x64_r = philox_engine<uint_fast64_t, 64, 2, r, 0xD2B74407B1CE6E93,
0x9E3779B97F4A7C15>;
```

- 2 Required behavior:** The 10000th consecutive invocation of a default-constructed object of type `philox2x64_r<10>` produces the value XXXXXXXXX

```
using philox2x32 = philox2x32_r<10>;
```

```
using philox2x64 = philox2x64_r<10>;
```

`philox4x32` and `philox4x64` define the most broadly used Philox parameter sets (supported in Intel® MKL, rocRAND, cuRAND, MATLAB, etc.).

`philox2x32` and `philox2x64` show good statistical properties and performance as well [8], but they are not broadly used across libraries.

Having two sets of aliases defined in the standard will complicate the choice and we decided to stick with the current consensus across the libraries by removing `philox2x32` and `philox2x64`.

[P2075R2](#) paper revision contained additional aliases, which allowed setting the number of rounds of the algorithm:

```
template<std::size_t r>
using philox4x32_r<r> = see below;
```

```
template<std::size_t r>
using philox4x64_r<r> = see below;
```

Theoretically the `philoxNxW_r<r>` permits the program to trade speed for safety by specifying a number of rounds of mixing. Philox generators with `r=7` have no known statistical flaws [2].

But in practice only rounds equal to 10 are widely used and implemented in a variety of the libraries. Taking that into consideration we simplified the user's choice by removing these additional aliases. The experts, who seek for finer control over statistical properties, can define such aliases in their code.

8 Counter X and its sequence representation

Counter X behaves as a big integer but is represented in some parts of the algorithm description and in `set_counter()` function as a sequence of values X_i . Users need to understand which words in this sequence represent the higher order parts of the mathematical X value in order to use the `set_counter()` function in a meaningful way (see section [set counter use case](#)). That controls, whether they should write:

```
eng.set_counter({atomid, timestep, 0, 0});
```

or

```
eng.set_counter({0, 0, timestep, atomid});
```

for the Philox period to be evenly split between all computational blocks of the program.

In order to define the API but leave implementation details in hands of implementers, we specify the order of this only in the description of `set_counter()` function behavior.

9 Vectorization

Philox can be vectorized on ARM and x86 architectures:

- A. Philox algorithm is defined to generate a batch of n values, which can be vectorized. Additional optimization opportunities exist for generating several batches for several counter values X at once (though it may increase the state to store intermediate results).
- B. Philox's round steps require the next simple math operations: multiplication, addition and XOR. Depending on the optimization scheme, shuffle / permutation operations may be also required.

All required vectorization instructions exist in all modern architectures thus the schema is equally applicable.

9 Polls

LEWG Telecon on 2023-11-28

POLL: Which philox architecture do we prefer?

1 engine template (6. Philox-Focused)	Neutral	engine template + prf template (7. Generic counter_based_engine)
8	2	0

10 Impact on the Standard

This is a library-only extension. It adds one or two new class templates, zero or one new concepts, and a small number of pre-defined template aliases.

11 References

- 1 P1932R0 "Extension of the C++ random number generators":
<http://open-std.org/JTC1/SC22/WG21/docs/papers/2019/p1932r0.pdf>.
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